

Titan System Architecture Overview

Purpose

This document provides the complete architectural overview of TitanZero.

It summarises how the platform is structured and how its major components interact.

This document acts as the master blueprint for:

developers

system architects

AI agents

extension builders

It describes the complete system stack from user interfaces to database architecture.

Titan Architecture Philosophy

TitanZero is designed as an event-driven, AI-governed business operating system.

The platform combines:

lifecycle-driven business processes

signal-based automation

AI-assisted decision support

modular extensions

The system architecture separates responsibilities into five layers:

Interfaces

Assistants

Governance

Core Engines
Data Layer

Each layer has clearly defined responsibilities.

Layer 1 — Interfaces

Interfaces are how users interact with the platform.

Primary interfaces include:

Titan Command
Titan Talk
Titan Origin
Titan Go

Each interface serves a specific role.

Titan Command

Business control centre dashboard.

Titan Talk

Unified communication interface.

Titan Origin

Traditional administrative interface.

Titan Go

Mobile/PWA interface for field operations.

Interfaces interact with the system through engine APIs.

They never access database tables directly.

Layer 2 — Assistants

Assistants provide intelligent support for business processes.

Two assistant families exist.

LogiCore Assistants
CreatiCore Assistants

LogiCore Assistants

Focus on operational lifecycles.

Examples:

job lifecycle assistants

dispatch assistants

finance lifecycle assistants

quality control assistants

CreatiCore Assistants

Focus on marketing and creative workflows.

Examples:

campaign assistants

content generation assistants

engagement optimisation assistants

Assistants monitor system signals and generate action proposals.

Layer 3 — Governance

Governance is managed by the Titan Zero Engine.

Titan Zero acts as the system's AI oversight layer.

It ensures that important decisions follow defined policies.

Key responsibilities:

assistant orchestration

AI model routing

decision evaluation

approval workflows

Titan Zero uses a trinary reasoning model.

OmegaCore

OmicronCore

TitanZeroCore

OmegaCore

Strategic reasoning and planning.

OmicronCore

Detailed analysis and validation.

TitanZeroCore

Decision synthesis and governance.

For high-risk actions Titan Zero activates Trinary Bayesian Mode.

Layer 4 — Core Engines

Core engines implement the fundamental capabilities of the platform.

Primary engines include:

- Titan Identity Engine
- Titan Customer Engine
- Titan Work Engine
- Titan Evidence Engine
- Titan Docs Engine
- Titan Money Engine
- Titan Talk Engine
- Titan Signal Engine
- Titan Pulse Engine
- Titan Zero Engine
- Titan Analytics Engine

Each engine manages a specific domain of functionality.

Engines expose APIs used by interfaces, assistants, and automation.

Layer 5 — Data Layer

The data layer stores all system information.

Tables follow a strict prefix doctrine.

- tz_* operational business data
- ext_* extension module data
- (no prefix) platform system tables

Examples:

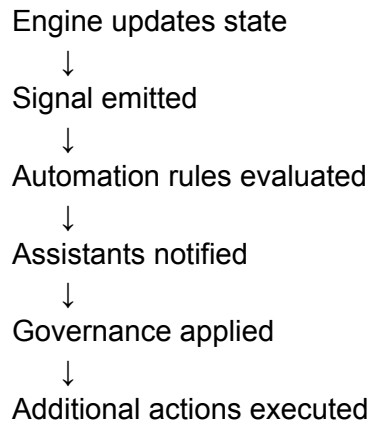
- tz_jobs
- tz_customers
- tz_invoices
- tz_evidence_items
- ext_social_campaigns
- users
- teams

The data layer supports multi-tenant operation using team_id.

Event-Driven Architecture

TitanZero uses an event-driven model.

Core flow:



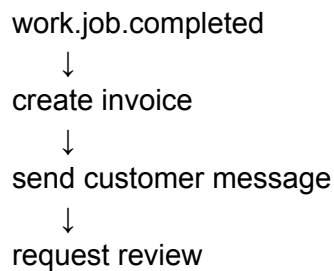
This architecture allows the system to react dynamically to business events.

Automation Layer

Automation is executed by the Titan Pulse Engine.

Pulse listens to signals and executes automation rules.

Example automation:



Automation allows the platform to operate efficiently with minimal manual intervention.

AI Integration

AI functionality is managed through the Titan Zero Engine.

Titan Zero provides:

assistant orchestration

AI reasoning

model routing

artifact generation

governance oversight

AI models are selected through the system's multi-modal routing architecture, which chooses the most appropriate model for each task.

Signal Bus

The Titan Signal Engine acts as the event bus for the entire system.

Every meaningful system action emits a signal.

Examples:

work.job.created

finance.invoice.sent

talk.message.received

evidence.uploaded

Signals allow automation, assistants, and analytics to respond to system activity.

Analytics Layer

Analytics is provided by the Titan Analytics Engine.

It processes signals to generate metrics and insights.

Examples include:

operational dashboards

financial reporting

performance analytics

Analytics provides decision-making insight for business operators.

Extension Ecosystem

The platform supports a modular extension ecosystem.

Extensions allow developers to add new capabilities without modifying core engines.

Examples:

marketing automation modules

AI content tools

external integrations

industry-specific modules

Extensions use the ext_* table prefix and integrate through signals and engine APIs.

PWA Platform

TitanZero operates as a Progressive Web Application.

The platform supports:

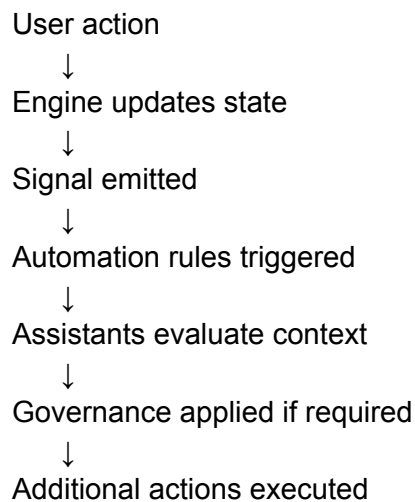
desktop browser

mobile browser
installed PWA mode

This allows the system to behave like a native mobile application while remaining web-based.

Titan Operational Loop

The complete system operation can be summarised as:



This loop forms the operational core of the Titan platform.

Summary

TitanZero is an event-driven, AI-assisted business operating system.

By combining lifecycle-driven data models, signal-based automation, AI assistants, and governance oversight, the platform enables businesses to operate with greater efficiency and intelligence.

The architecture is designed to be modular, scalable, and extensible, allowing the platform to evolve while maintaining stability and operational reliability.